Experiment 2: STOICHIOMETRY OF SOLUTIONS: DETERMINATION OF ACETIC ACID CONTENT IN VINEGAR

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Abstract

This study determined the acetic acid content in a commercial sample of Marca Piña "Sukang Puti" vinegar using acid-base titration to test whether the experimental mass percentage matched the manufacturer's declared 4.5% concentration. A 5.00-mL vinegar sample was titrated against a standardized 0.2129 M sodium hydroxide (NaOH) solution utilizing a phenolphthalein indicator. Over three trials, with burette volumes intended to be recorded to the nearest 0.01 mL to minimize computational error, the experimental acetic acid concentrations were calculated as 2.072%, 2.141%, and 2.118%, yielding an average of 2.110% (w/w). This calculated average falls significantly below the declared 4.5% label value, a discrepancy attributed to potential errors in sample measurement, burette reading, and visual endpoint estimation. Despite this deviation from the expected label value, the consistent results across the three replicate titrations indicate that the fundamental laboratory procedures were executed reliably.

Keywords: Acetic acid, acid-base titration, stoichiometry, sodium hydroxide, mass percentage, Marca Piña.

1. Introduction

Vinegar is a standard commercial solution composed primarily of acetic acid, and its precise concentration can be quantified through acid-base titration, an analytical technique that relies on the 1:1 stoichiometric neutralization between an acid and a base. While commercial labels declare a specific concentration, such as the 4.5% acetic acid claim on the tested Marca Piña "Sukang Puti" sample, potential inconsistencies in production or degradation over time raise the question of whether the actual chemical composition matches the manufacturer's specifications. To investigate this, the primary objective of this experiment is to determine the true mass percentage of acetic acid in the commercial vinegar sample by neutralizing it with a standardized 0.2129 M sodium hydroxide (NaOH) solution in the presence of a phenolphthalein indicator. By utilizing the exact stoichiometric molar ratio between acetic acid and sodium hydroxide to perform concentration calculations, this study tests the hypothesis that the experimentally determined percentage will align with the 4.5% label claim, simultaneously assessing the precision of laboratory titration methods across three replicate trials.

2. Methodology

2.1. Burette Preparation

A 25-mL burette was thoroughly cleaned using a mild laboratory detergent and a burette brush, after which it was rinsed with tap water to remove traces of the used detergent. The burette was then rinsed out using approximately 10 mL of distilled water, allowing the water to drain completely through the stopcock after each rinse. The inside of the burette was inspected carefully to ensure that the water drained uniformly without droplets sticking to the glass surface. The stopcock was also checked for proper operation and possible leakage before proceeding with the titration.

2.2. Conditioning and Filling of the Burette

The burette was conditioned by rinsing it with approximately 5 mL of the standardized 0.2129 M sodium hydroxide (NaOH) solution, ensuring that the solution came in contact with the entire inner surface. A small amount of NaOH was then allowed to flow through the stopcock and burette tip to remove any remaining distilled water inside. Any trapped air bubbles in the burette tip were removed by allowing additional NaOH solution to flow through the tip. The burette was then filled with fresh 0.2129 M NaOH solution using a clean funnel, after which the funnel was removed. The burette was securely clamped to an iron stand in a vertical position, and any liquid droplet hanging from the burette tip was removed by gently touching the tip against the inside wall of the waste beaker. The initial burette reading was recorded to the nearest 0.01 mL by observing the bottom of the meniscus at eye level.

2.3. Titration of the Vinegar Sample

Using a 10-mL graduated cylinder, 5.00 mL of "Vinegar A (Marca Piña 'Sukang Puti')" was measured and transferred into a clean 250-mL Erlenmeyer flask. Approximately 20 mL of the distilled water was added, followed by two drops of phenolphthalein indicator. The solution was gently swirled to obtain a homogeneous mixture.

The Erlenmeyer flask was positioned beneath the burette, and the standardized 0.2129 M NaOH solution was gradually added while the flask was being continuously swirled. As the endpoint was approached, the NaOH solution was added one drop at a time while swirling after each addition. The endpoint was considered reached when a

faint pink color persisted throughout the solution for at least 30 seconds. The final burette reading was recorded to the nearest 0.01 mL, and the volume of NaOH used was determined from the difference between the initial and final burette readings.

The titration procedure was then repeated using a fresh 5.00-mL vinegar sample for a total of three trials. The volumes of NaOH consumed in the three trials were used to determine the amount of acetic acid present, and the calculated percent-by-mass values were averaged to obtain the experimental acetic acid content of the vinegar.

2.4. Determination of Acetic Acid Content

The amount of sodium hydroxide used during each titration was calculated from its standardized molarity and measured volume. Based on the 1:1 stoichiometric relationship between sodium hydroxide and acetic acid in the neutralization reaction,

$$CH_3COOH(aq) + NaOH(aq) \rightarrow CH_3COONa(aq) + H_2O(l)$$

the number of moles of acetic acid present in each vinegar sample was taken to be equal to the number of moles of NaOH consumed.

The mass of acetic acid was subsequently calculated using its molar mass of 60.05 g/mol. The mass of each vinegar sample was determined using the 5.00-mL sample volume and the given vinegar density of 1.049 g/mL. The percent by mass (% w/w) of acetic acid was then determined from the ratio of the calculated mass of acetic acid to the mass of the vinegar sample. The percent-by-mass values obtained from the three trials were averaged and compared with the declared acetic acid concentration of 4.5% indicated on the vinegar label.

2.5. Waste Disposal and Cleanup

After the titration was completed, the neutralized titration mixtures were disposed of through the laboratory sink with sufficient running water in accordance with the prescribed laboratory waste-disposal procedure. Two drops of phenolphthalein indicator were added to the beaker containing the remaining sodium hydroxide rinse solution. The solution was then neutralized by adding 1.0 M hydrochloric acid (HCl) dropwise while stirring until the pink color disappeared, indicating neutralization. The neutralized solution was disposed of through the sink with running water as instructed. All glassware were thoroughly rinsed with distilled water, the work area was cleaned, and the apparatus was returned to the laboratory storage location.

3. Results and Discussion

3.1 Experimental Data

The vinegar sample was analyzed through acid-base titration using 0.2129 M NaOH. A 5.00 mL vinegar sample was used for each of the three trials, and the volume of NaOH needed to reach the endpoint was recorded. The experimental data and calculated results are shown in Table 1.

Table 3.1.1 Experimental Results for the Determination of Acetic Acid in Vinegar

Parameter	Trial1	Trial 2	Trial 3
Initial burette reading (mL)	25.00	25.00	25.00
Final burette reading (mL)	16.50	16.20	16.30
Volume of NaOH used (mL)	8.50	8.80	8.70
Moles of NaOH (mol)	0.00181	0.00187	0.00185
Moles of acetic acid (mol)	0.00181	0.00187	0.00185
Mass of acetic acid (g)	0.1087	0.1123	0.1109
Volume of vinegar sample (mL)	5.00	5.00	5.00
Mass of vinegar sample (g)	5.245	5.245	5.245
Acetic acid, % (w/w)	2.072%	2.141%	2.118%

3.2 Moles of Sodium Hydroxide and Acetic Acid

The titration was based on the following balanced chemical equation:

$HC_2H_3O_2 + NaOH \rightarrow NaC_2H_3O_2 + H_2O$

The reaction has a 1:1 mole ratio between acetic acid and sodium hydroxide. Therefore, the moles of NaOH used in each trial were equivalent to the moles of acetic acid present in the vinegar sample.

For Trial 1, the moles of NaOH were calculated using its molarity and volume:

Moles of NaOH = Molarity × Volume

= (0.2129 mol/L)(0.0085 L)

= 0.00181 mol NaOH

Since the reaction has a 1:1 mole ratio:

0.00181 mol NaOH = 0.00181 mol $HC_2H_3O_2$

The same procedure was applied to Trials 2 and 3, resulting in 0.00187 mol and 0.00185 mol of acetic acid, respectively.

3.3 Mass of Acetic Acid

The moles of acetic acid obtained from each trial were converted to mass using the molar mass of acetic acid, which is approximately 60.05 g/mol.

For Trial 1:

Mass of acetic acid = Moles × Molar mass

= (0.00181 mol)(60.05 g/mol)

≈ 0.1087 g

The calculated masses of acetic acid for Trials 2 and 3 were 0.1123 g and 0.1109 g, respectively.

3.4 Percentage by Weight of Acetic Acid

The percentage by weight of acetic acid was calculated by comparing the mass of acetic acid with the mass of the vinegar sample.

% (w/w) Acetic Acid = (Mass of Acetic Acid / Mass of Vinegar Sample) × 100

Using the calculated values, the percentages of acetic acid were 2.072% for Trial 1, 2.141% for Trial 2, and 2.118% for Trial 3.

3.5 Average Percentage of Acetic Acid

The average percentage of acetic acid was determined using the results from all three trials:

Average % (w/w) = (2.072% + 2.141% + 2.118%) / 3

= 2.110% (w/w)

Thus, the average experimental concentration of acetic acid in the vinegar sample was 2.110% (w/w).

Conclusion

The sample of vinegar was titrated to determine the acetic acid content, using acid base titration and 0.2129 M NaOH. Based on the three trials, the acetic acid concentrations were 2.072%, 2.140%, and 2.118% and an average of 2.110% (w/w). This value was less than the 4.5% on the label of the vinegar but this can be explained by the errors in measuring the sample, reading the burette and estimating the endpoint of the titration. The three titrations gave similar results which indicated the titrations were carried out reliably.

References

[1] Chemistry Cluser, “Experiment 1: Basic Laboratory Techniques and Operations.” in *General Chemistry Laboratory Manual (CM011L)*. Manila, Philippines: School of Chemical, Biological, and Materials Engineering and Sciences, Mapua University, 2026. pp. 36-44.

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Criteria for Grading Lab Reports

Overall (OA): 5 points
Does it contain the required sections?
Is it clearly written?
Does it use good grammar and scientific terms properly?
Are the words spelled correctly?
Are the calculations performed correctly?
Does the title page contain the author's name and address?

Abstract (A): 5 points
Does it introduce the overall topic?
Does it explain the hypothesis being tested?
Are the important methods described?
Does it reach logical conclusions supported by the data?
Does it flow well? Is it logically written? Is it concise?

Introduction (I): 10 points
Does it give general background?
Does it point out poorly understood or unknown factors related to the study?
Does it raise questions?
Does it explain the hypothesis being tested? Does it contain the objectives of the experiment?
Does it flow well? Is it logically written? Is it concise?

Materials and Methods (M): 5 points
Could the experiments be understood based on the information given?
Does it include the source of the reagents?
Does it describe all of the methods used?
Is it excessively long?

Results and Discussion (R&D): 20 points
Does it explain the rationale and strategy for the experiments performed?
Does it describe, in words, what was done?
Does it summarize the findings obtained in the Results section?
Does it flow well? Is it logically written? Is it concise?
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Figures and Tables: Are the figures and tables well designed?
Do the figures include informative legends?
Do the figures present information useful for understanding the text?

Conclusion (C): 5 points
Does it answer the questions raised in the Introduction?
Does it answer the objectives contained in the Introduction?
Does it reach conclusions?
Does it explain why the conclusions are important?